


12/9/22

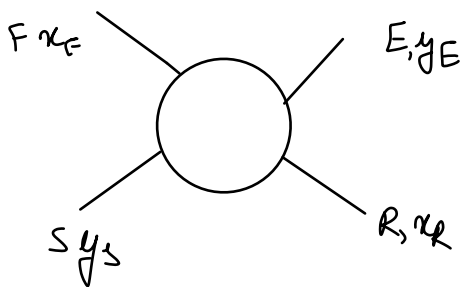
$$\rightarrow x_F = 0.28$$

Q 40 kg of feed with 28% solute is mixed with 30 kg of solvent that has 2% carrier in it. A single stage batch extractor.

Find the amount and composition of the extract & raffinate

→ At least 5 points to be plotted on paper/graph in exams.

→
Always
make this
also



Feed = carrier +
solute

$A + C + B \approx B \rightarrow \text{raffinate}$
 $B + C + B \approx A \rightarrow \text{extract}$

$$F + S = 70 \text{ kg}$$

$$F x_F + S y_S = E$$

→ You have to use approximation in graph

→ Two columns of two data set each

→ First two columns for raffinate

→ As soon as you get pair of points, join the tie line,
don't wait!!

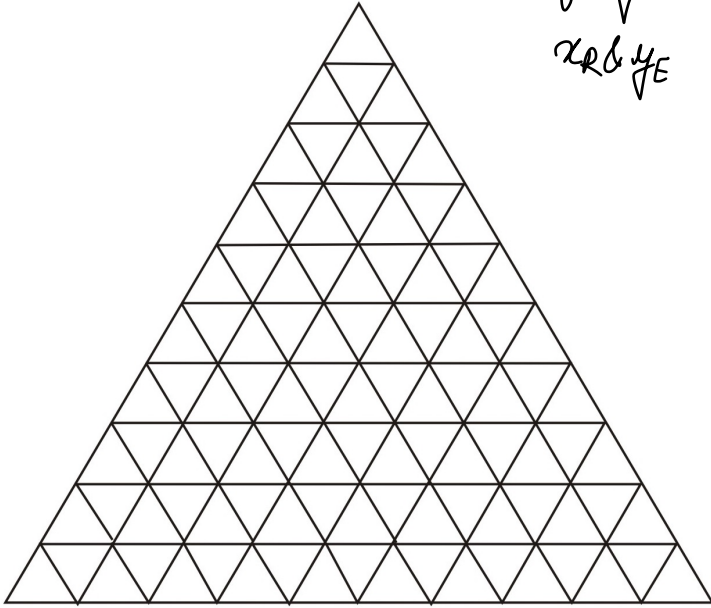
→ $S + F = M$ $M = 70$

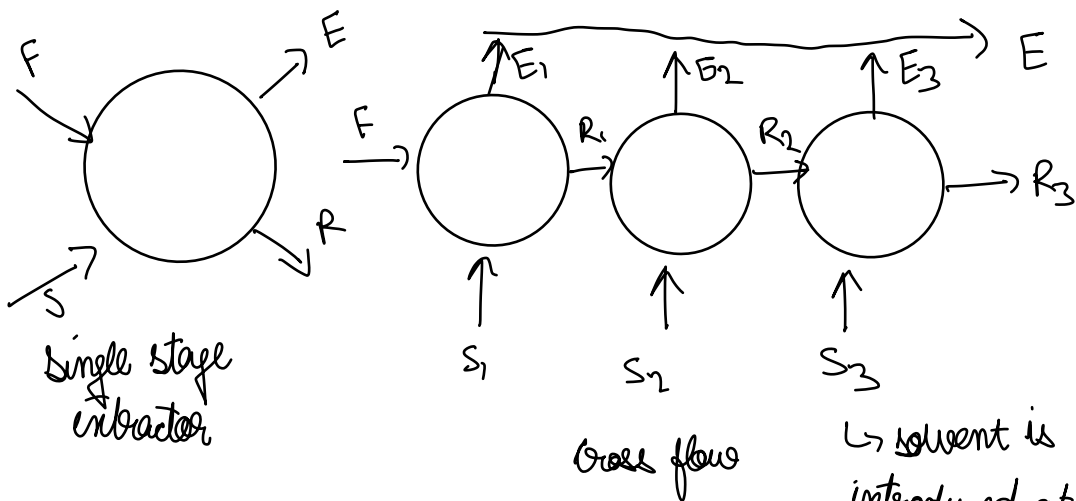
$S y_S + F x_F = M x_M$

$40 \times 0.28 + 30 \times 0.02 = 70 \times x_M$

$x_M = 0.168$

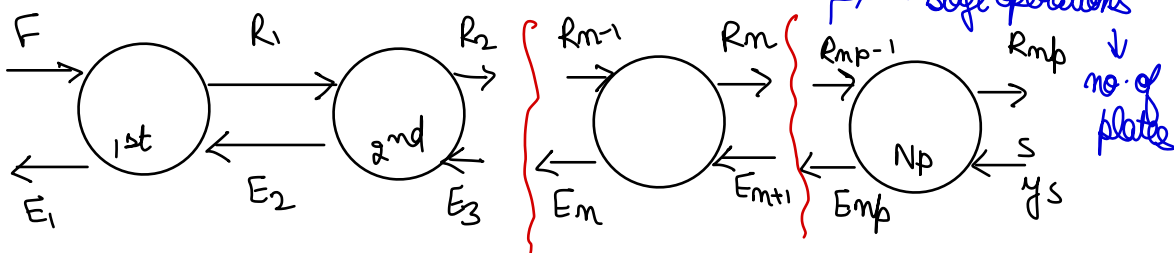
→ after plotting M ,
check the tie line passing through
 M .
→ and ends of tie lines become
 x_B & y_E





solvent is introduced at every stage

Design of counter flow is easy



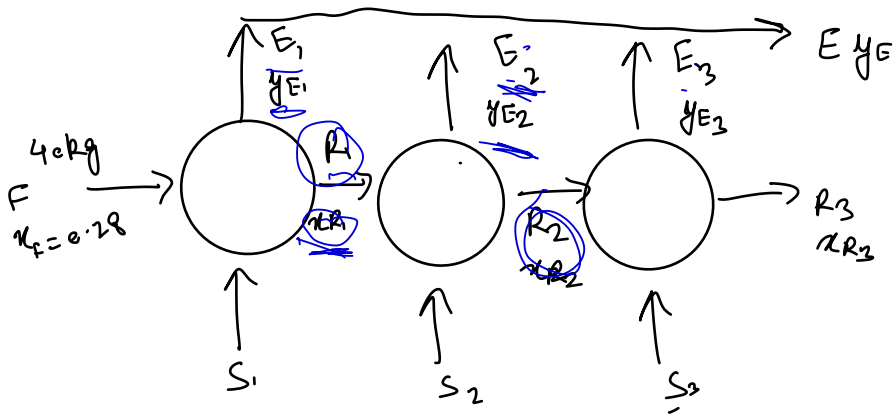
Counter flow → solvent is introduced at last

Why multi-stage?

to improve efficiency stage

Because some C may be present in raffinate and we don't want to waste it.

Q 40 kg 28% C $\rightarrow$ feed, cross flow configuration with 3 stages, extracted with a solvent which contains 2% A as contamination, solvent flow is 15 kg for each stage. Final extract quality and amount. Raffinate amount, Raffinate composition



$$M x_m =$$

$$55.7 \times x_m = E_2 \times y_{E_2} + R_2 \times x_{R_2}$$

$$x_m = 0.124$$

$$R_2 = 31.2$$

$$E_2 = 24.5$$

Solution

$$M_1 = 55$$

$$F + S_1 = M_1$$

First stage

$$F \alpha_F + S_1 \alpha_S = M_1 \alpha_{m1}$$

$$40 \times 0.28 = 55 \times \alpha_{m1}$$

$$\alpha_{m1} = 0.2036$$

$$\mu_{E1} = 0.31$$

$$\alpha_{R1} = 0.16$$

$$E = 18 \quad R_1 = 37$$

Second stage

$$R_1 + S_2 = M_2$$

$$37 + 15 = M_2$$

$$R_1 \alpha_{F1} + S_2 \alpha_S = M_2 \alpha_{m2}$$

$$\alpha_{m2} = 0.10$$

$$55.7 \times 10^3 = E_2 y_{E_2} + R_2 x_{R_2}$$

$$\eta_{E_2} = 0.18$$

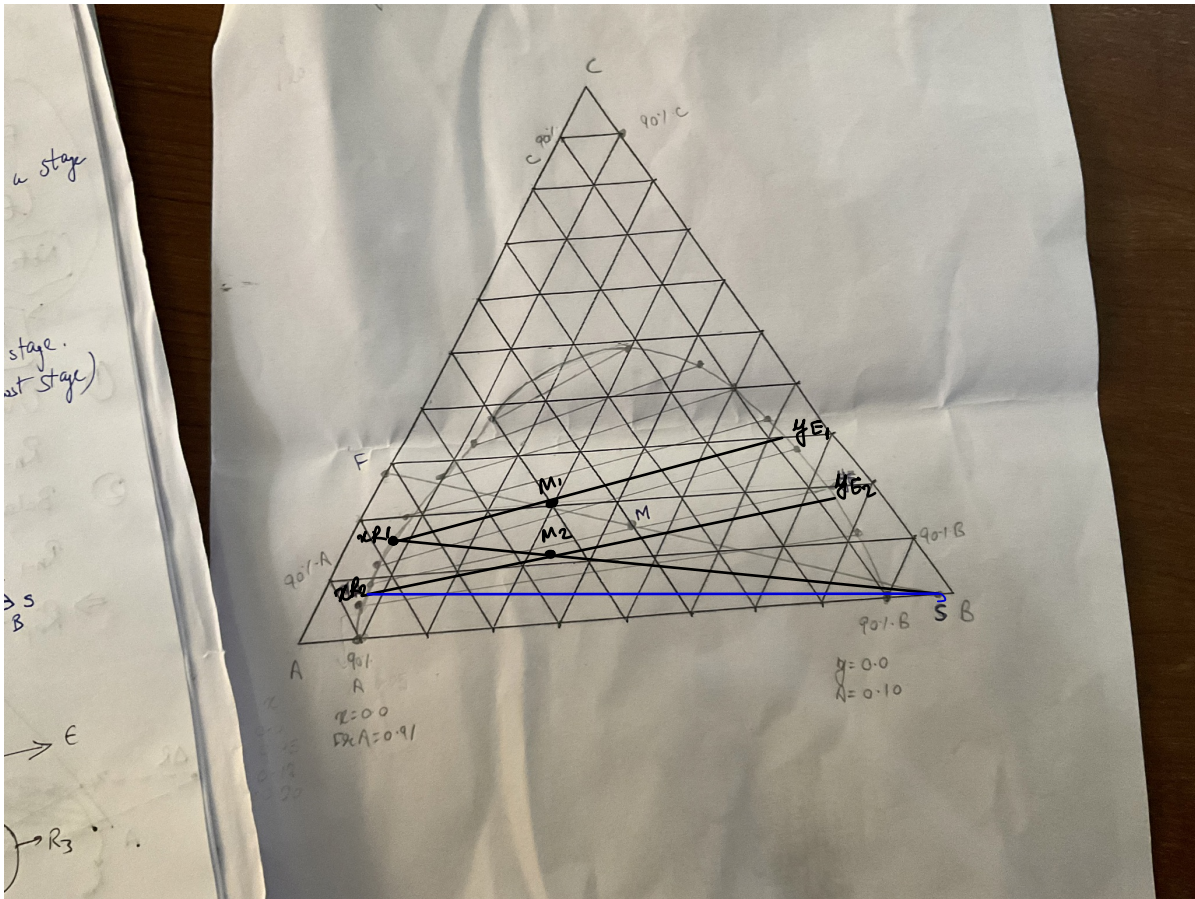
$$u_{R2} = 0.8$$

$$6.90 = E_2 \times 0.18 + R_2 \times 0.8 \quad 6.9 : (557 - R) \times 0.18 + R \times 0.8$$

$$55.7 = E_2 + R_2$$

$$SS-7 = E_2 + R_2$$

$$0 \cdot 1R \cong R =$$



Overall balance

$$F + S = E_1 + R_{NP} = M \Rightarrow R_{NP} - S = F - E_1 = \Delta R \quad \text{--- (1)}$$

$$\underline{F} \underline{x}_F + \underline{S} \underline{y}_S = \underline{E} \underline{y}_1 + \underline{R}_{NP} \underline{x}_{NP} = \underline{M} \underline{x}_m$$

Note: (E, γ_E) or (R_{NP}, γ_{NP}) should be defined

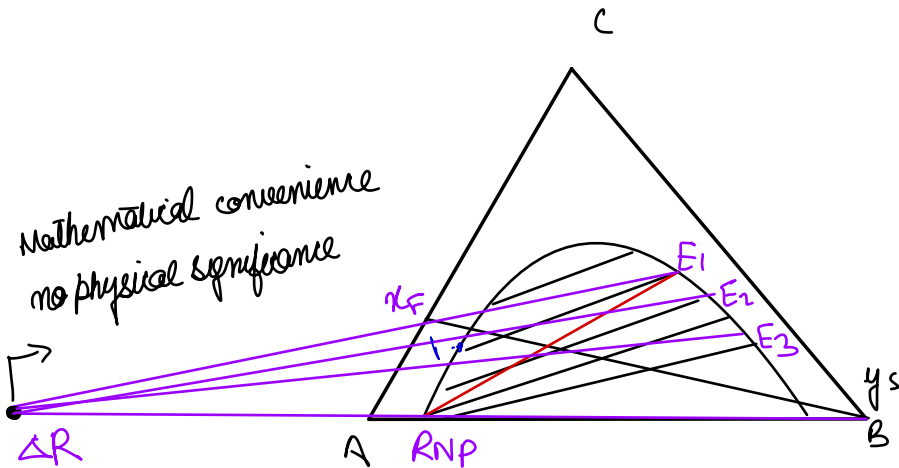
→ m for a counter flow is just a mathematical quantity, no real life significance.

i) Overall balance for stage n

$$R_{N-1} + E_{n+1} = R_n + E_n \Rightarrow R_{n-1} - E_n = R_n - E_{n+1} = \Delta R$$

2) Balance for whole plant from m^{th} stage to N^{th} stage

$$R_{m-1} + S = E_m + R_{NP} \Rightarrow R_{NP} - S = R_{m-1} - E_m = \Delta R \quad [\text{from ①}]$$



We do not draw the tie line through M
M will still lie on E_1 & R_{NP} line

$S - R_{NP} = E_1 - F = -\Delta R \rightarrow$ so a line joining E_1 & F goes to ΔR

For second stage $F - E_1 = \Delta R$

$$R_1 - E_2 = \Delta R$$

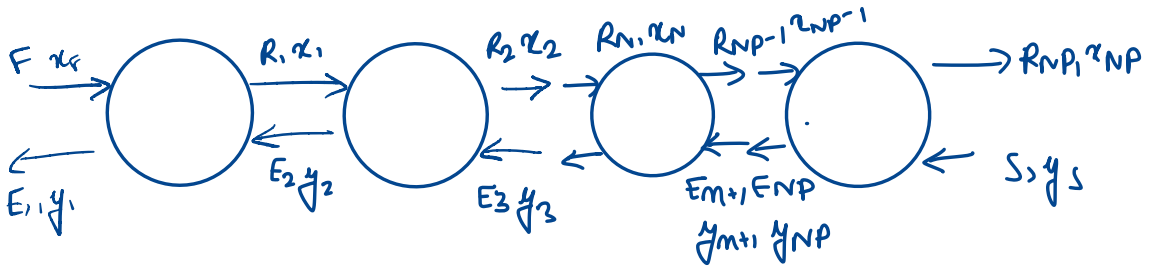
$$R_2 - E_3 = \Delta R$$

$\rightarrow$ First we get ΔR , using R_{NP} , S & E_1, F

$\rightarrow$ Now using tie line $E_1 - R_1$, we will locate R_1 and then
join to ΔR and extend to get E_2 , now using $E_2 - R_2$
tie line we will get R_2 and so on....

Note $\div$ why not use mathematics to solve $\rightarrow$ because we don't
have equilibrium eqns of each step.

Continuous counter current multistage extractor



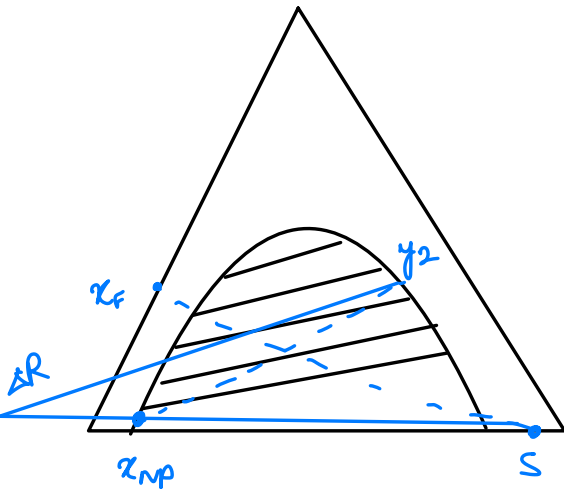
Overall balance

$$F + S = E_1 + R_{NP}$$

$$E_1 - F = S - R_{NP} = \Delta R$$

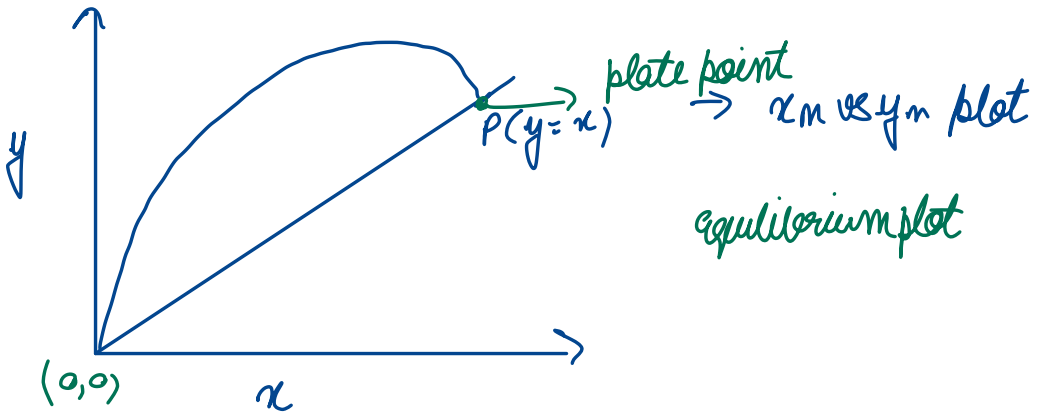
For n^{th} to NP^{th} stage : $R_{N-1} + S = E_N + R_{NP}$

$$E_N - R_{m-1} = S - R_{NP} = \Delta R$$



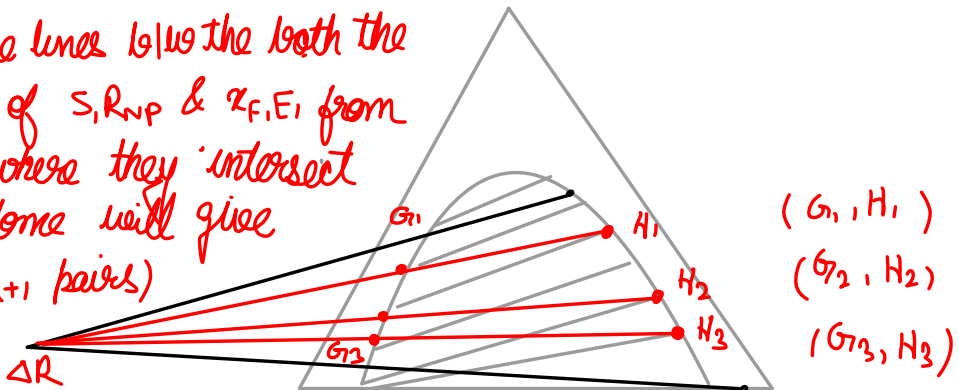
① x_m vs y_m plot $\rightarrow$ equilibrium plot

② x_m vs y_{m+1} plot $\rightarrow ?? \rightarrow$



all the lines b/w the both the
corners of S, R_{NP} & $x_{F,E}$ from
 ΔR where they intersect
the dome will give

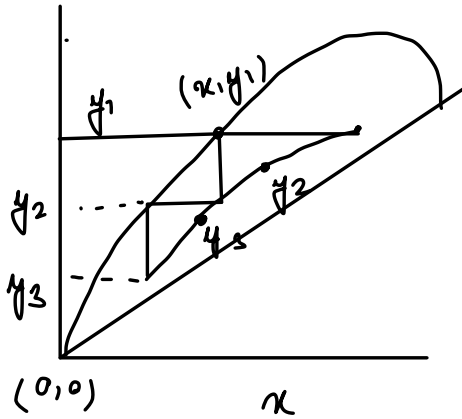
(x_m, y_{m+1}) pairs)



x_{G_1}, y_{G_1}
 $\hookrightarrow x_N \quad \hookrightarrow y_{m+1}$

For exam draw 4 lines
so you will 6 pts for
 x_m, y_{m+1} plot

y_1 vs x_F



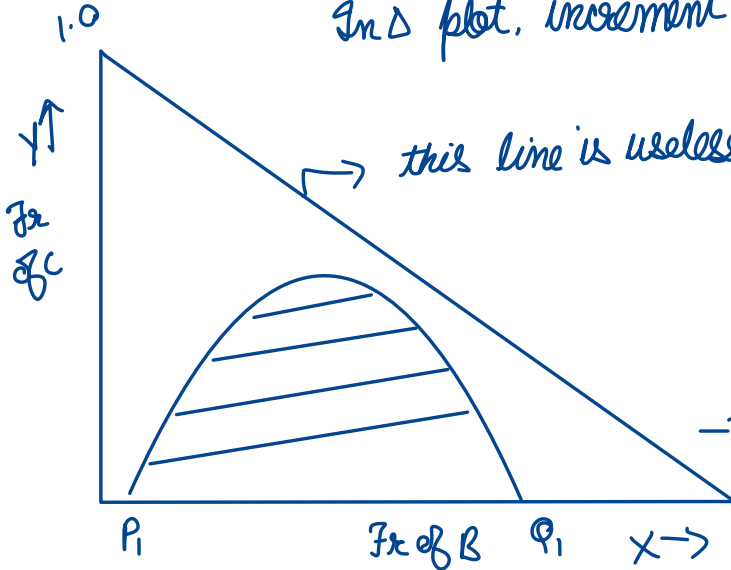
fraction of C $\leftarrow x$ fraction of B $\rightarrow$ fraction of C
 y Fr. of B

0	0.09	0	0.9
0.06	0.07	0.12	0.80

$P_1 (0, 0.09)$
 $Q_1 (0, 0.9)$

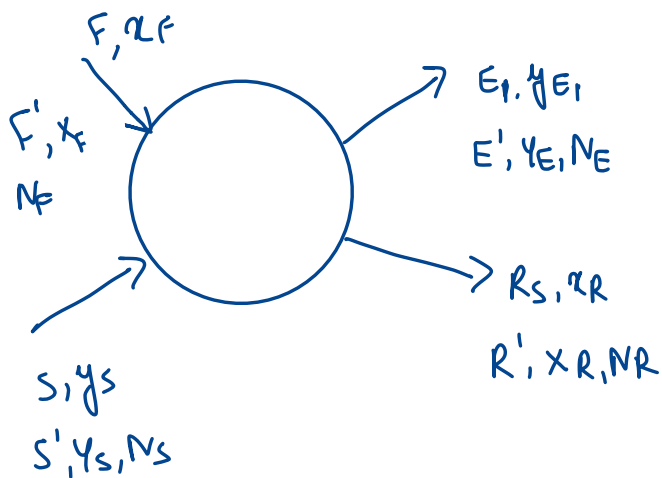
equilibrium pts

In Δ plot, increment is 10:1.



this line is useless, use it just to make it look like triangular plot

This is drawn on a graph paper where increment can be of 1:1.



$$x_F = \frac{C_F}{A_F + B_F + C_F}$$

$$y_E = \frac{C_E}{A_E + B_E + C_E}$$

$$x_F = \frac{C_F}{A_F + C_F}$$

$$y_E = \frac{C_E}{A_E + C_E}$$

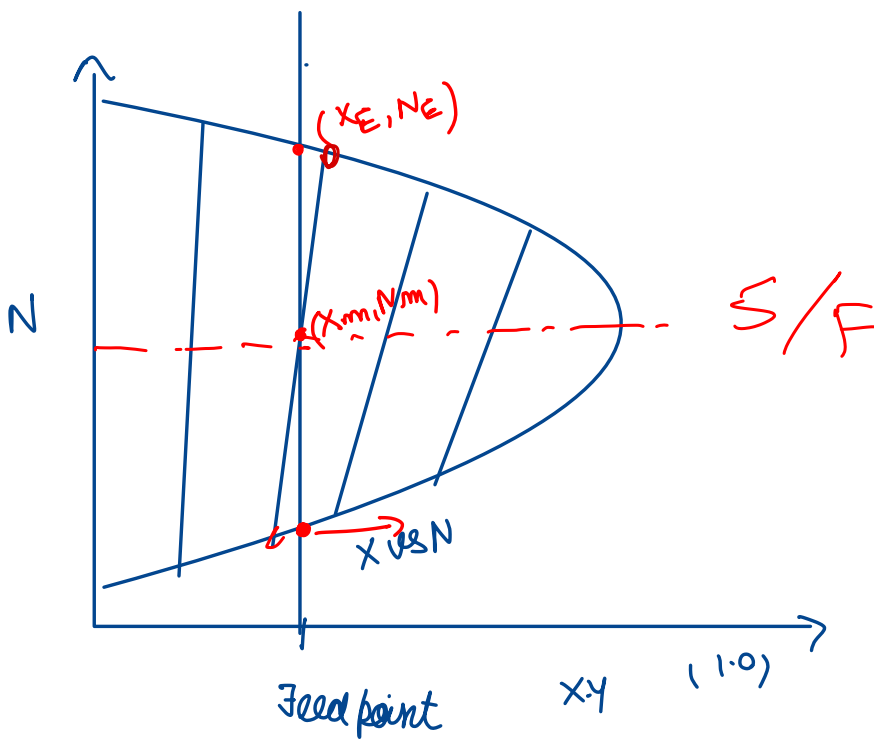
$$N_F = \frac{B_F}{A_F + C_F}$$

$$N_E = \frac{B_E}{A_E + B_E}$$

$$\left(x_S = \frac{C_S}{A_S + C_S} = \frac{0}{0} \right)$$

$$\left(N_S = \frac{B_S}{A_S + C_S} = \frac{B_S}{0} \right)$$

↳ In case of pure solvent



$$F = (A_F + B_F + C_F)$$

$$F' = (A_F + C_F) \rightarrow \text{actually this is not a physical quantity}$$

B Free overall balance

$$F' + \cancel{S'} = M' = E' + R'$$

for pure solvent $S' = 0$

$$F = A_F + C_F + B_F$$

$$F' = A_F + C_F = F$$

$$F' = M = F$$

B free C balance

$$F'X_F + S'Y_S = M'X_m$$

$$F'X_F = (\cancel{A_F} + C_F) \frac{C_F}{(\cancel{A_F} + \cancel{C_F})}$$

$= C_F \rightarrow C$ entering from feed

$$F'X_F = M'X_m$$

$$X_F = X_m$$

F, M, S are collinear

$$N_m = \frac{B_m}{A_m + C_m}$$

$$B_m = B_s + B_f \rightarrow 0$$

$$A_m = A_s \rightarrow 0 + A_f$$

$$C_m = C_s \rightarrow 0 + C_f$$

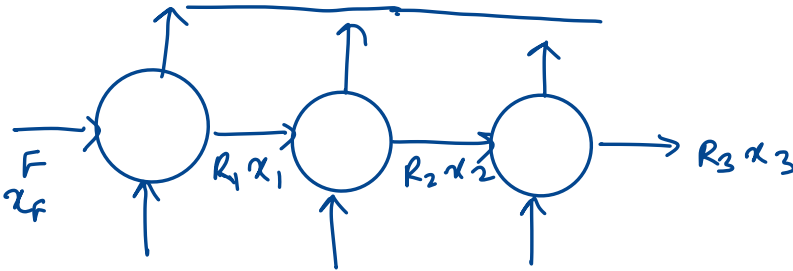
$$E = E' (1 + N_E)$$

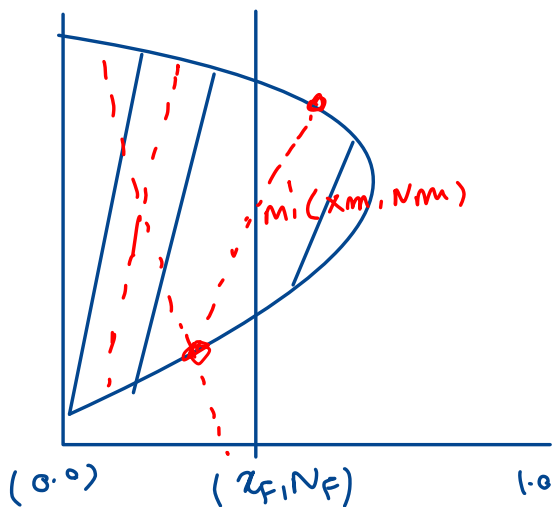
$$R = R' (1 + N_R)$$

$$E' = A_E + C_E$$

$$1 + N_E = 1 + \frac{B_E}{A_E + C_E} = \frac{A_E + B_E + C_E}{A_E + C_E}$$

Cross flow by B free Basis (Pure solvent)





we need to calculate N_m for each stage as solvent flow rate for each stage can be different.

Counter flow on B Free basis

Overall balance $F + S = M = E + R_{NP} \rightarrow$ summation of

Overall B free balance $F' + S' = M' = E' + R_{NP}'$

$\hookrightarrow$ summation of two components (A, C) that are entering

all three comp. - one that are entering

$$S' = 0$$

$F' = M'$ — For pure solvent $F = F'$

B free C balance

$$F'x_F + S'y_S \xrightarrow{C} M'x_m = E'y_E + R_{NP}'x_{NP}$$

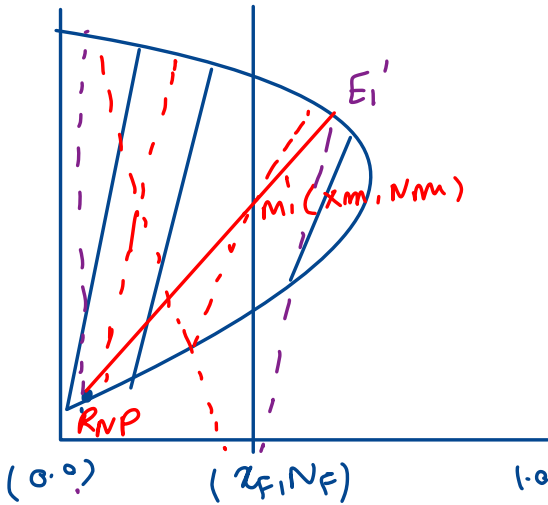
$$\Rightarrow x_F = x_m$$

B Free B balance

$$F'N_F + S'N_S = M'N_M$$

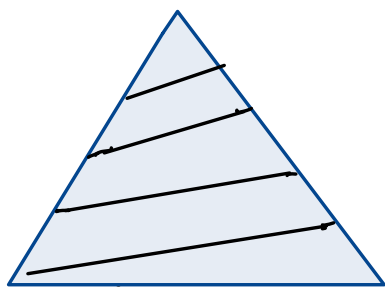
$$S'N_S = (A_S + C_S) \frac{B_S}{(A_S + C_S)} = B_S = S$$

↳ although $A_S + C_S = 0$



OR

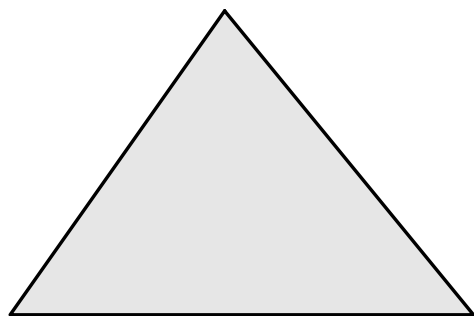
27/10/22



Phase diagram

where A and B are completely immiscible

→ It would be complete triangle with not horizontal tie-lines



Phase diagram

where A, B, C are completely miscible

→ Complete triangle

We can use "c" free notation → only when A and B

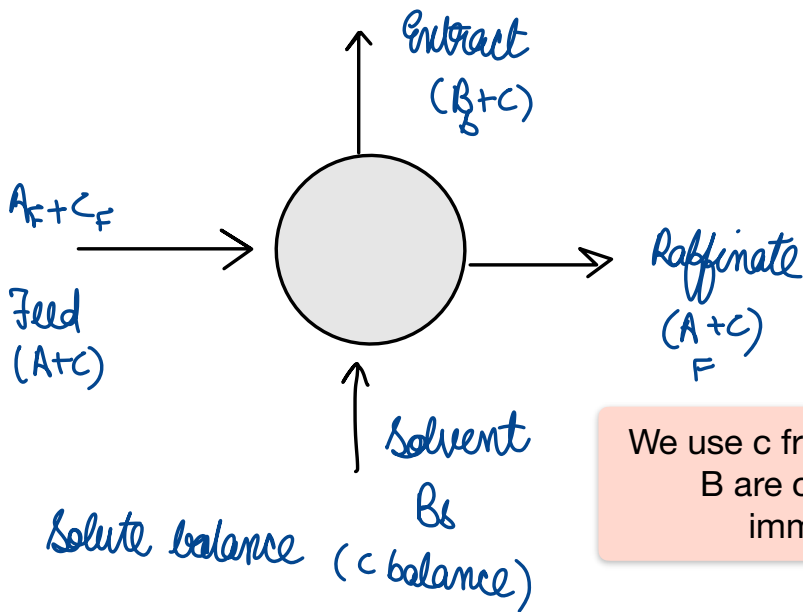
↳ this can be used only in this condition → are completely immiscible
"B" free → can be used anywhere

$$x' = \frac{x}{1-x}$$

$$y' = \frac{y}{1-y}$$

x → stream of raffinate (including feed)

y → stream of extract (including solvent)



$$A_F x_F' + B_S y_S' = A_R x_R' + B_E y_E'$$

$$\left(A_F \cdot \frac{C_F}{A_F + B_F} \right) + B_S \cdot \frac{C_S = 0}{B_S + A_S} = A_R \frac{C_R}{A_S + B_S} + B_E \cdot \frac{C_E}{B_E + A_E}$$

(pure solvent)

because A, B completely immiscible

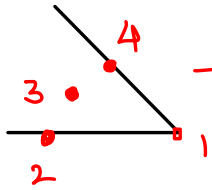
$$\left. \begin{array}{l} A_R = A_F = A \\ B_S = B_E = B \end{array} \right\} \text{as A, B completely immiscible}$$

$$A x_F' + B y_S' = A x_R' + B y_E'$$

$$B(y_S' - y_E') = -A(x_F' - x_R')$$

$$\frac{y_S' - y_E'}{x_F' - x_R'} = -\frac{A}{B}$$

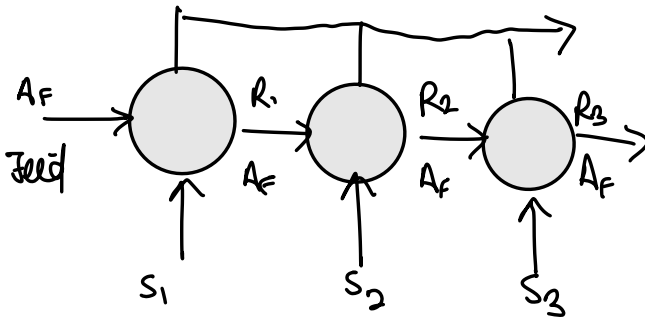
→ becomes important when equilibrium data in x', y' form is given



→ For completely immiscible A, B

2, 3 is not possible in solvent completely immiscible

→ as A, B



→ In completely immiscible A, B A_F would remain same

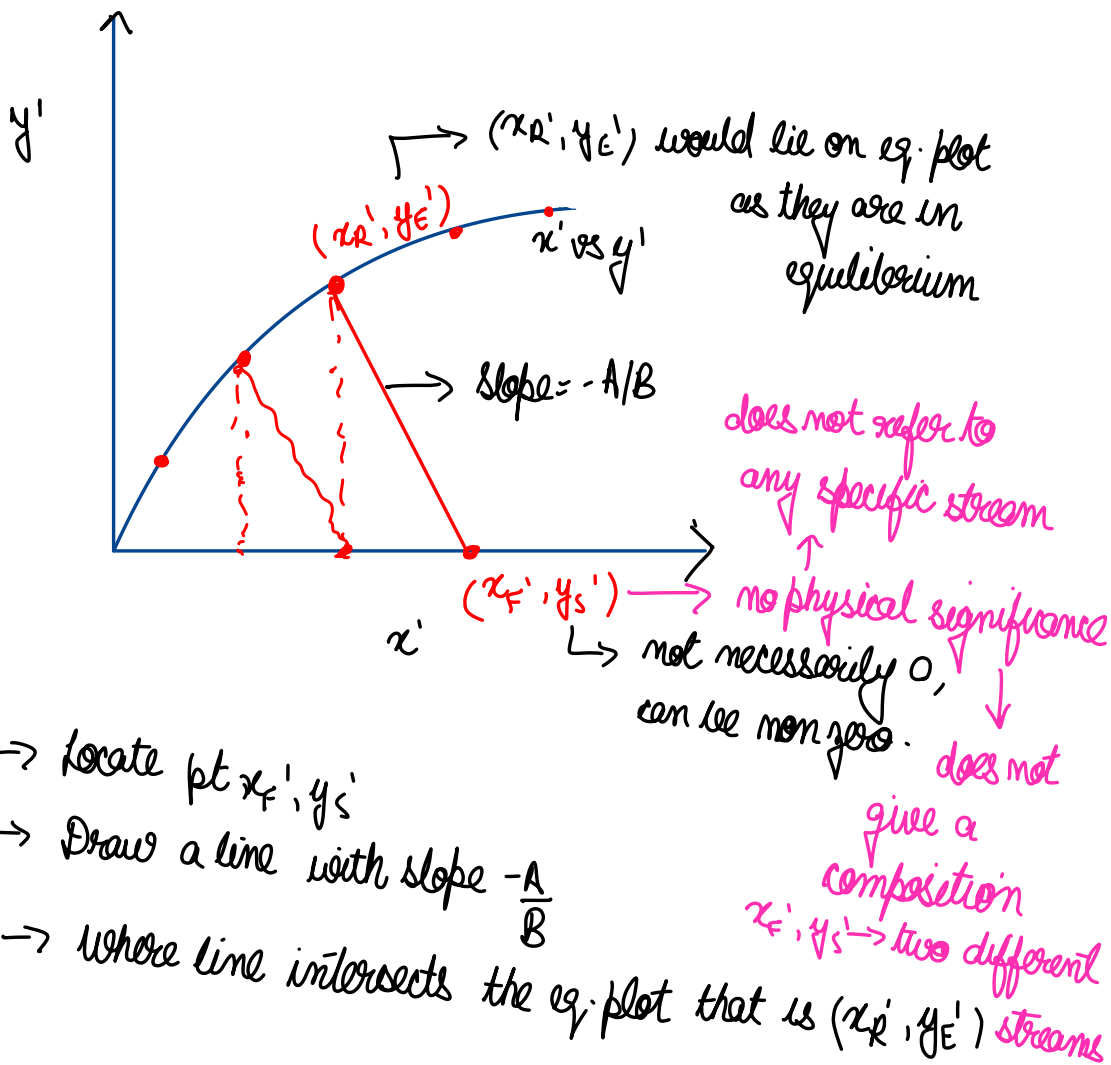
→ x_F' is known

→ Pure solvent is given

→ A, B is known

Disclaimers

$$\frac{y_S' - y_E'}{x_F' - x_R'} = -\frac{A}{B}$$

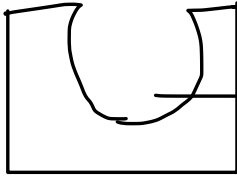


$\rightarrow$ locate pt x_F', y_S'

$\rightarrow$ Draw a line with slope $-\frac{A}{B}$

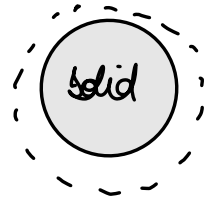
$\rightarrow$ Where line intersects the eq. plot that is (x_R', y_E')

Leaching



Solid

↳ Matrix / Host → must be solid



Solvent

Liquid

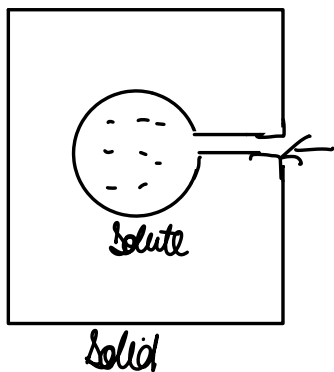
Leaching or Leaching

Decoction → when solvent is at boiling point

↳ It is a special case of leaching when solvent is at boiling point

Elutriation → when solute is mostly on the surface of the solid

Solution mining



① Diffusion of solvent to / through pores

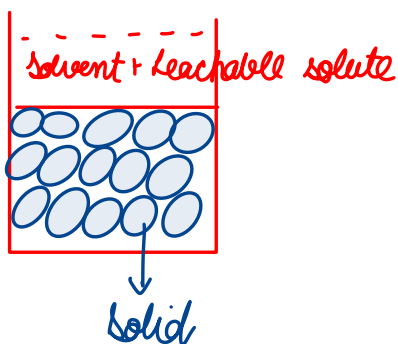
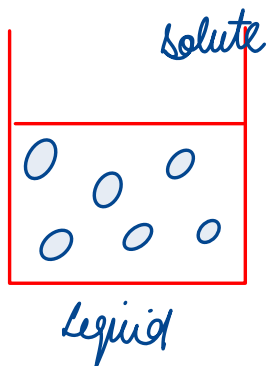
② Dissolution of the solute inside the pores

③ Outward diffusion (due to conc. gradient)

Not for exams

② Is every material leachable, provided that the solvent dissolves the solute

→ No, not every solute which is dissolved by solvent is not leachable, as the solvent should be able to penetrate



31/10/2022

1. Nicotine (C) in water (A) solution containing 1.0 % nicotine is to be extracted with kerosene (B). Water and Kerosene are completely insoluble. Determine the percentage extraction of nicotine if 100 kg of feed solution is extracted in (a) One single stage with 150 Kg solvent; (b) three stages in cross flow configuration. All the stages use 50 Kg of kerosene.

(C) Same feed is now extracted in a counter current system. The target is to reduce the Nikotene content to 0.1%. Determine the number of stages if 115 KG of kerosene is used.

Equilibrium data:

$\rightarrow 1.011 \times 10^{-3} \quad 2.46 \times 10^{-3} \quad 5.02 \times 10^{-3} \quad 7.51 \times 10^{-3} \quad 9.98 \times 10^{-3} \quad 20.4 \times 10^{-3}$

x' (kg nicotine / kg water)	0	0.001011	0.00246	0.00502	0.00751	0.00998	0.0204
y^* (kg nicotine / kg kerosene)	0	0.000807	0.001961	0.00456	0.00686	0.00913	0.0187

c_{free}
 $c_{fraction}$
 $x' = \frac{x}{1-x}$

a) $x_F = 0.01 \quad x_F' = \frac{0.01}{0.99} = 0.010101 \dots \quad y_S' = 0 \rightarrow \text{pure solvent}$

$F = 100 \text{ kg} \quad A = A_F = 99 \text{ kg} \quad B = 150 \text{ kg}$

① locate (x_F', y_S')

② $-\frac{A}{B} = -\frac{99}{150} = -0.66$

Now challenge is to draw line with desired slope

$\rightarrow$ so lets take $x_i' = 0.006$; y_i' $\rightarrow$ will find out

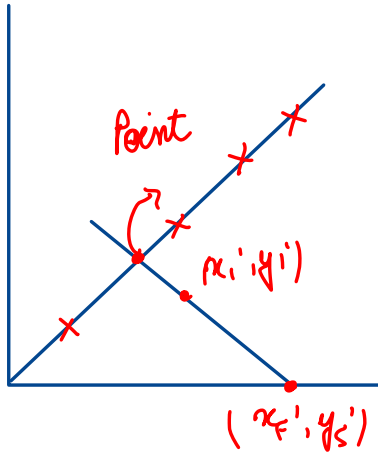
$$\frac{y_1' - y_5'}{x_1' - x_5'} = -0.66 \rightarrow \text{get } y_1'$$

plot x_1', y_1'

Join x_1', y_1' & x_5', y_5'

extend and let it intersect to eq. plot.

$$\hookrightarrow \text{ans}(0.0042, 0.00381)$$



absolute value of

c extracted

$$A_F(x_F' - x_R') \Rightarrow C_{inF} - C_{inR}$$

$\Downarrow$

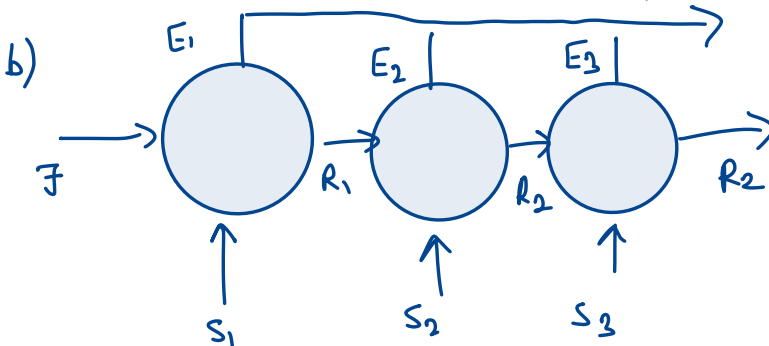
$$A_F \left(\frac{C_{inF}}{A_{inF}} - \frac{C_{inR}}{A_{inR}} \right)$$

same denominator

$\Downarrow$

$$\therefore c \text{ extracted} = \frac{A_F(x_F' - x_R')}{\text{Initial amount of } c \text{ present}} \times 100 = 58.41\% \text{ in this case}$$

$\hookrightarrow 1 \text{ kg in this case}$



$$S_1 = S_2 = S_3 = 50 \text{ kg}$$

$$x_F = 0.01$$

$$F = 100 \text{ kg}$$

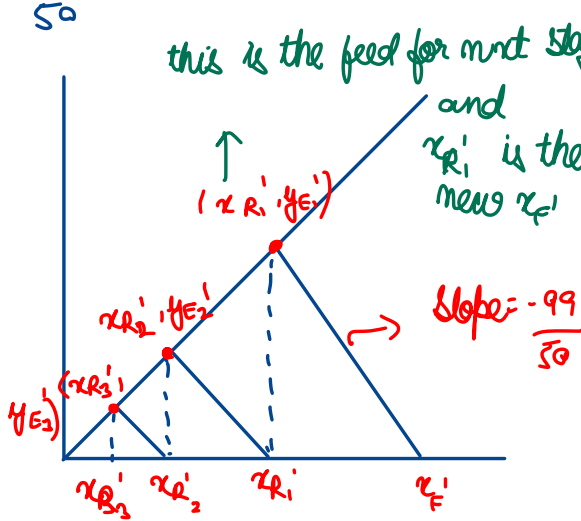
$$A_F = 99 \text{ kg}$$

$$x_F' = 0.0101$$

$$y_S' = 0$$

$$x \quad \frac{y_S' - y_E'}{x_F' - x_R'} = -\frac{A}{B}$$

$$-\frac{A}{B} = \frac{-99}{50} = -1.98$$



this is the feed for next stage, so drop down and x_{R1}' is the new x_F'

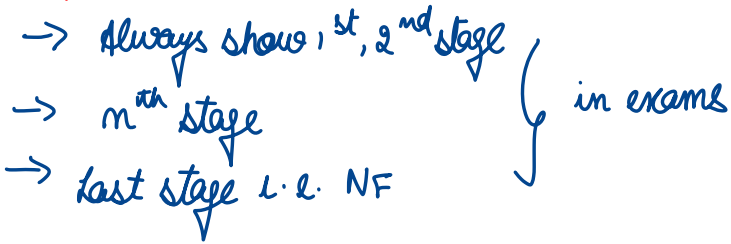
Treat stage 1 like a single stage

Then for stage 2 here the solvent B is same so we are getting same slope = $-\frac{A}{B}$

Disclaimer: Different values of B_1, B_2, B_3 in each stage could be given, so slope would change acc. $-\frac{A}{B_1}, -\frac{A}{B_2}, -\frac{A}{B_3}$ and soon [solvent flow rate for each stage is different]

A is same as A & B are completely immiscible

b) Ans $\rightarrow 67.1\%$



$$B y_5' + A x_F' = A x_{NP}' + B y_1' \quad \text{--- c balance on c free basis}$$

$$\frac{A}{B} = \frac{y_1' - y_2'}{x_1' - x_2'} \rightarrow \frac{A}{B} \text{ has +ve slope here}$$

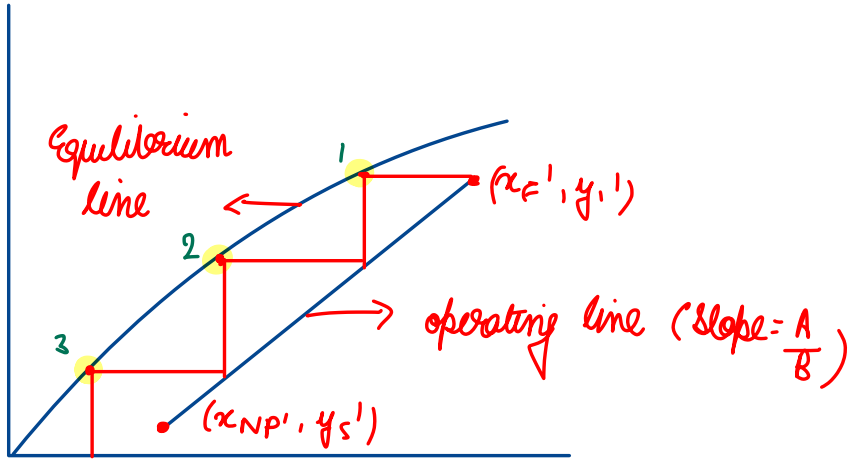
not -ve slope

$y_i \rightarrow$ will get, but we want no. of stages

$$y_1' = 0.0078$$

→ Find y_1'

→ locate x_{N+1}', y_s'
 x_F', y_1'



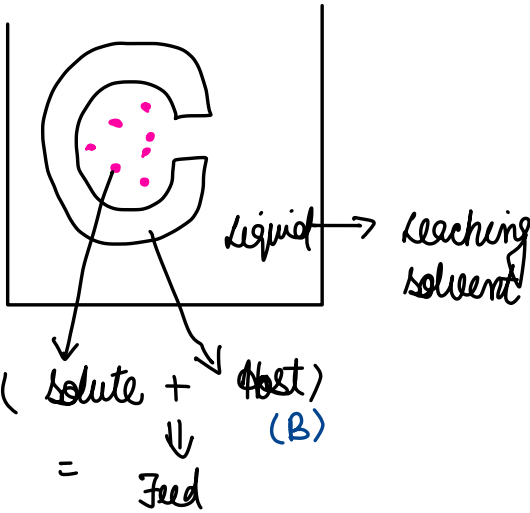
eq stage → the two composition
E & R are in equilibrium

operating line → locus of composition that can be possible
b/w two stages → locus of composition b/w two stages

The whole step is not the stage, the point where it intersects / touches the eq. curve is the actual stage

so in exam mark → intersection points as stage

Leaching



B → insoluble

A → solvent

C → solute

Different Notations for leaching

→ Standard

$$N = \frac{\text{Mass of Insoluble} \rightarrow B}{\text{Mass of (A+C)}}$$

stream

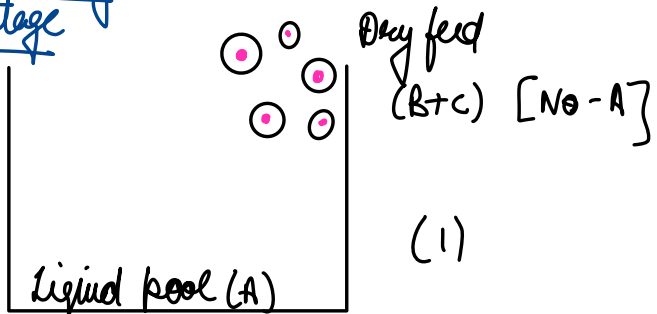
→ B-free base

x, y

$$x, y = \frac{\text{Mass of C}}{\text{Mass of (A+C)}}$$

stream

Leaching stage



→ (Diffusion + settling)

An ideal stage of leaching contains these two :-

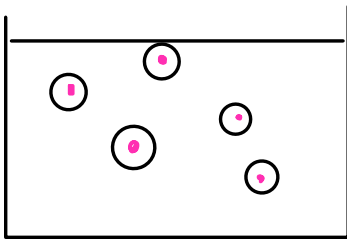
- Chuck out all solute from particles
- Will also lead to settling of insoluble

In extraction there was no defined preference whether extract or raffinate will be on top

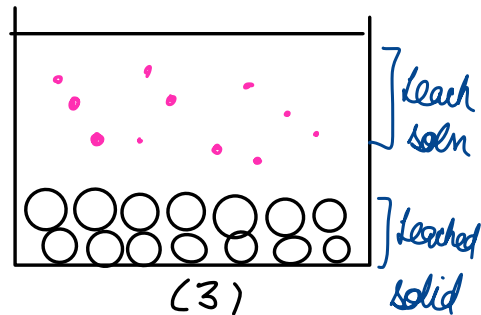
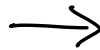
- Leached soln will go to top
 - Insoluble settles down
- } →

Here we talk about time as compared to extraction

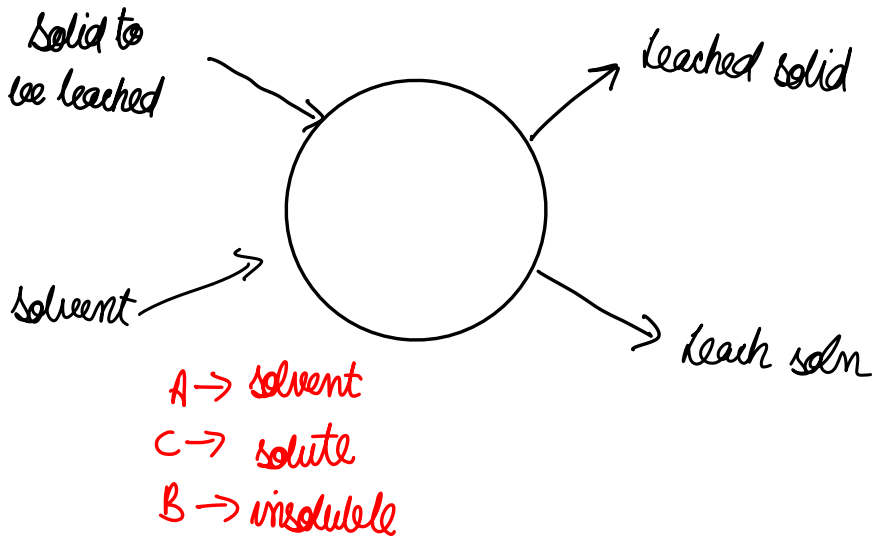
Diffusion of solid in liquid $\ll$ Diffusion of liquid in liquid



(2)



(3)



Solid to be leached

$B = \text{mass of insoluble}$

$$F = (A + C)$$

For dry feed $A = 0$

$$N_F = \frac{B_F}{(A+C)_F} = \frac{B_F}{C_F}$$

$$y_F = \frac{C_F}{(A+C)_F} = \frac{C_F}{C_F} = 1.0$$

$\downarrow$
 in leaching y_F is used
 for feed, in extraction x_F was used

Solvent

R_0 : mass of the stream

$$x_0 = \frac{C / \text{stream}}{(A+C) / \text{stream}} = 0$$

$$N_0 = \frac{B / \text{stream}}{(A+C) / \text{stream}} = 0$$

Solid to be Leached

$B = \text{Mass of Insolubles}$

$F = (A+C)$ For Dry Feed $A=0$

$$N_F = \frac{B_F}{(A+C)_F} = \frac{B_F}{C_F} = (\text{Some Value})$$

$$y_F = \frac{C_F}{(A+C)_F} = \frac{C_F}{C_F} = 1.0$$

Solvent

R_0 , mass of the stream

$$x_0 = \frac{C|_{\text{stream}}}{(A+C)|_{\text{stream}}} = 0$$

$$N_0 = \frac{B|_{\text{stream}}}{(A+C)|_{\text{stream}}} = 0$$

Leached Solid

$B = \text{Mass of Insoluble}$

$E_1 = \text{Mass of } (A+C)$

$$N_{E_1} = \frac{B}{E_1}$$

$$y_1 = \frac{C+E}{E_1}$$

Both undefined

Leach Soln

$R_1 = \text{Mass of Soln}$

$$x_1 = \frac{\text{mass of } C}{(A+C)|_R} = \text{Some Value}$$

$$N_{R_1} = \frac{B|R}{(A+C)|_R} = 0$$

Leached Solid

B = Mass of insoluble

E_1 = Mass of $(A+C)$

$$N = \frac{B}{E_1} \rightarrow \infty$$

$$y_1 = \frac{C \ln E}{E_1} = 0/0$$

Both
undefined
can't be
plotted on
graph

Leach Soln

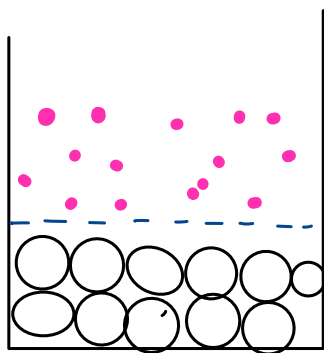
R_1 = mass of soln

$$x_1 = \frac{\text{mass of } C}{(A+C)_R} \rightarrow \text{some finite value}$$

$$NR_1 = \frac{B \ln R}{(A+C)_R} = 0$$

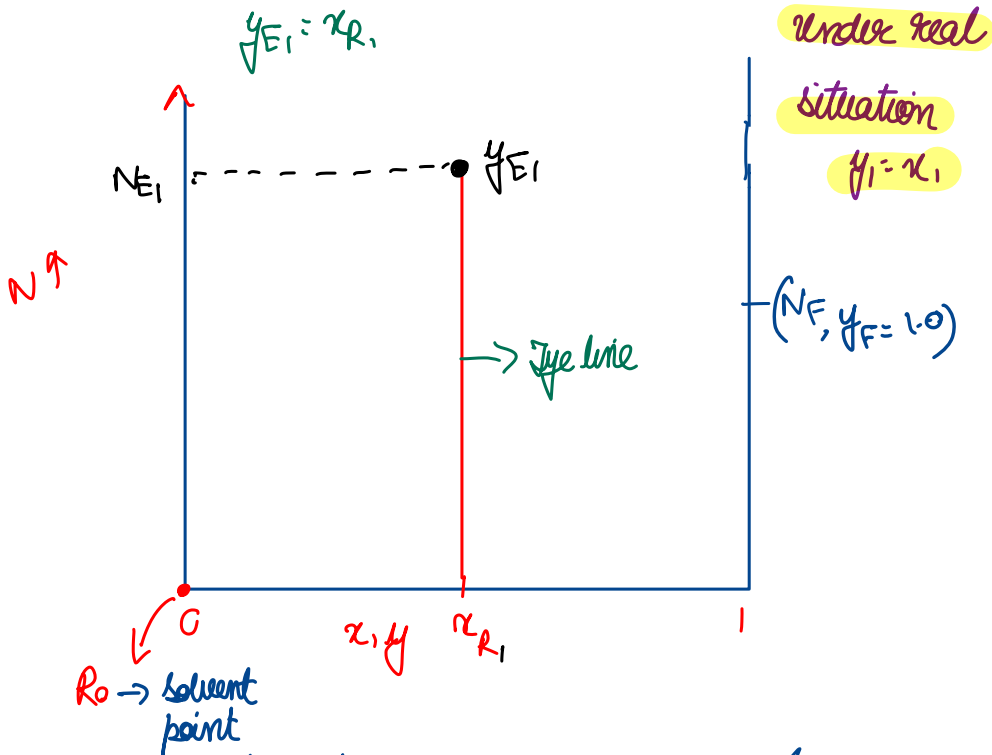
Solid is liquid free and liquid is solid free $\rightarrow$ Ideal stage with 100% efficiency

$\downarrow$
our assumption



$\rightarrow$ Leached solid only contains solid $\rightarrow$ ASSUMPTION!!!
 $\rightarrow$ assumption that interested spaces are more void is wrong $\rightarrow$ Physically impossible situation

Reality $\rightarrow$ particles will be surrounded by some liquid, and also the composition of liquid would be same as that of leached solution
 $\rightarrow$ Practical equilibrium



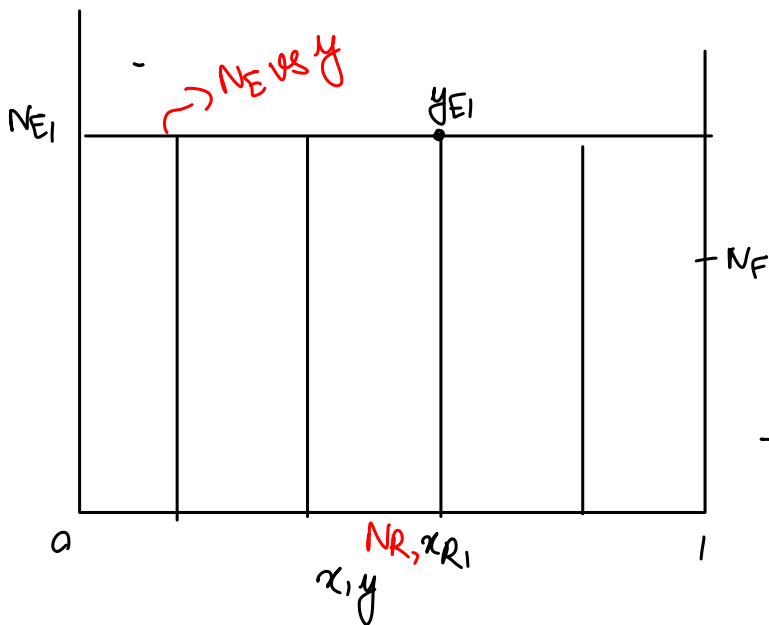
→ Percolated liquid has brought in some amount of A & C in leached solid

Tie line → line connecting two ^{phases} pt. which are in practical eq. with each other
 ↳ not theoretical

↳ Leached solid & leach soln are in eq. with each other

→ If size of particles are too small, they may not settle, and form a colloidal

→ If size of particles are too big / large, the solvent may not be able to leach out the solute.



→ All these vertical lines are different - different tie lines

→ For a given composition, tie line would be vertical

Consideration

1. Some solute unleached → this alters y_{EI} .
2. Some B may be dispersed in the liquid → this make $NR, \neq 0$
3. Surface re-adsorption of C on to B → this does not alter the y_{EI} .

$$y_{EI, \text{real}} = \frac{C + C_{\text{stuck}}}{A + C + C_{\text{stuck}}}$$

$C \rightarrow$ C inherited from percolated liquid

$$\frac{C}{A+C} = 1 - \frac{C}{A+C} = 1 - y$$

11.11.22

No slip boundary condition - Reason to justify practical equilibrium
↓

A liquid layer over a solid surface is attached to it
composition of liquid above solid is same as that below
solid layer

- For design we assume it to be ideal
- Liquid percolates and fills up the void
- Liquid above the solid & below the void has same composition

→ Assumptions → when we started

① No solid in R_1 →

② No liquid in E_1 → overwritten using practical equilibrium

↳ No solute

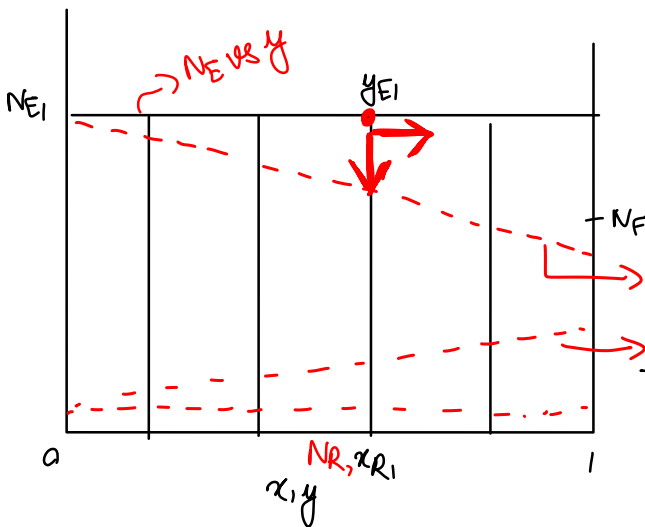
↳ No solvent →

No liquid percolated +
no solute in B. to percol
solid B at
the bottom

- Mass of insoluble (B) is same for every trial
- mass of solution / liquid percolated is same in every trial
 - ↳ some amount of (A+C)
- ↳ Reason why N_E vs y is parallel to x, y axis.
- Most likely, all solute has not come out from the settled insoluble B

① If there is some B in the leached soln: N_E vs x line does not coincide with x -axis.

↓
just shift, we can't predict whether it will be parallel, +ve, -ve slope, linear, non-linear



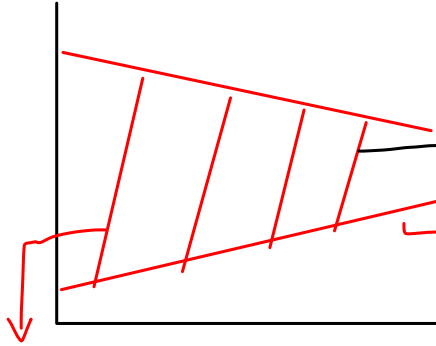
$N_E \rightarrow B \rightarrow \text{same}$
 $\bar{E}_1 \rightarrow (A+C) \rightarrow \text{increase}$
 so N_E decreases

→ shifting of N_E line
 → shifting of N_R line

→ N_E vs γ not // → C, A stuck in pores

→ $N_R \neq 0$ → some B is floating

→ some C in interstitial may be adsorbed too → but this doesn't affect the mathematics



→ updated N vs γ .

can't predict whether tie line would be slanted towards left or right

→ shifting of eq. pts along with N_R, N_E lines

slanted tie lines are not parallel

$N_E = B$ → B is reduced.

$A+C$ → $A+C$ is increasing

some B going to leached soln
C remaining stuck in B

N_E → pushed down

N_R → pushed up

→ only these two things are guaranteed

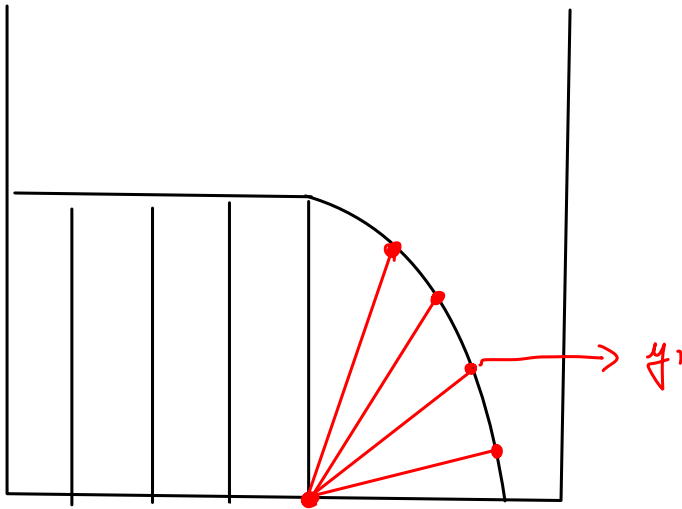
with time slopes will relax



time will also effect the graph.

→ diverge a bit

Solubility limit $\rightarrow$



$\rightarrow$ solubility limit

$\hookrightarrow$ beyond this more
conc. of solute can't
be present in liquid

